

OUTPUTS

3.1 Warm Up Exercises:

1. Sorting and Subsetting:

Complete all following Task:

- Dataset for the Task: "titanic.csv"

Following task is common for all the problem:

1. Load the provided dataset and import in pandas DataFrame.
2. Check info of the DataFrame and identify following:

Problem 1 - Sorting:

1. Create a DataFrame called fare that contains only the Fare column of the Titanic dataset. Print the head of the result.
2. Create a DataFrame called class age that contains only the Pclass and Age columns of the Titanic dataset, in that order. Print the head of the result.
3. Create a DataFrame called survived gender that contains the Survived and Sex columns of the Titanic dataset, in that order. Print the head of the result.

```
Fare DataFrame (head):
```

```
   Fare
0    7.2500
1   71.2833
2    7.9250
3   53.1000
4    8.0500
```

```
Class Age DataFrame (head):
```

```
   Pclass  Age
0        3  22.0
1        1  38.0
2        3  26.0
3        1  35.0
4        3  35.0
```

```
Survived Gender DataFrame (head):
```

```
   Survived  Sex
0         0  male
1         1 female
2         1 female
3         1 female
4         0  male
```

Problem - 2 - Subsetting:

Complete all the following Task:

Subsetting Rows:

1. Filter the Titanic dataset for cases where the passenger's fare is greater than 100, assigning it to fare_gt_100. View the printed result.
2. Filter the Titanic dataset for cases where the passenger's class (Pclass) is 1, assigning it to first_class. View the printed result.
3. Filter the Titanic dataset for cases where the passenger's age is less than 18 and the passenger is female (Sex is "female"), assigning it to female_under_18. View the printed result.

Passengers with fare>100:

	PassengerId	Survived	Pclass	\
27	28	0	1	
31	32	1	1	
88	89	1	1	
118	119	0	1	
195	196	1	1	

		Name	Sex	Age	SibSp	\
27		Fortune, Mr. Charles Alexander	male	19.0	3	
31	Spencer, Mrs. William Augustus (Marie Eugenie)		female	NaN	1	
88		Fortune, Miss. Mabel Helen	female	23.0	3	
118		Baxter, Mr. Quigg Edmond	male	24.0	0	
195		Lurette, Miss. Elise	female	58.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
27	2	19950	263.0000	C23 C25 C27	S
31	0	PC 17569	146.5208	B78	C
88	2	19950	263.0000	C23 C25 C27	S
118	1	PC 17558	247.5208	B58 B60	C
195	0	PC 17569	146.5208	B80	C

Passengers in first class:

	PassengerId	Survived	Pclass	\
1	2	1	1	
3	4	1	1	
6	7	0	1	
11	12	1	1	
23	24	1	1	

		Name	Sex	Age	SibSp	\
1	Cummings, Mrs. John Bradley (Florence Briggs Th...		female	38.0	1	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)		female	35.0	1	
6		McCarthy, Mr. Timothy J	male	54.0	0	
11		Bonnell, Miss. Elizabeth	female	58.0	0	
23		Sloper, Mr. William Thompson	male	28.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
1	0	PC 17599	71.2833	C85	C
3	0	113803	53.1000	C123	S
6	0	17463	51.8625	E46	S
11	0	113783	26.5500	C103	S

Subsetting Rows by Categorical variables:

1. Filter the Titanic dataset for passengers whose Embarked port is either "C" (Cherbourg) or "S" (Southampton), assigning the result to embarked c or s. View the printed result.
2. Filter the Titanic dataset for passengers whose Pclass is in the list [1, 2] (indicating first or second class), assigning the result to first second class. View the printed result.

Passengers embarked at Cherbourg (C) or Southampton (S):

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

Passengers in first or second class:

	PassengerId	Survived	Pclass	\
1	2	1	1	
3	4	1	1	
6	7	0	1	
9	10	1	2	
11	12	1	1	

	Name	Sex	Age	SibSp	\
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
6	McCarthy, Mr. Timothy J	male	54.0	0	
9	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	
11	Bonnell, Miss. Elizabeth	female	58.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
1	0	PC 17599	71.2833	C85	C
3	0	113803	53.1000	C123	S
6	0	17463	51.8625	E46	S
9	0	237736	30.0708	NaN	C

3.2 Exploratory Data Analysis Practice Exercise - 1.

Warning: Handle missing values in the Age column by filling them with the median age of the dataset before performing the division.)

Answer the following questions from Dataset:

Which passenger had the highest fare paid relative to their age?

To answer the question perform following operations:

1. Add a column to the Titanic dataset, fare per year, containing the fare divided by the age of the passenger(i.e., Fare/Age).
2. Subset rows where fare per year is higher than 5, assigning this to high fare age.
3. Sort high fare age by descending fare per year, assigning this to high fare age srt.
4. Select only the Name and fare per year columns of high fare age srt and save the result as result.
5. Look at the result.

```
Passengers with high fare per year:
      Name  Fare per Year
305  Allison, Master. Hudson Trevor    164.728261
297  Allison, Miss. Helen Loraine      75.775000
386  Goodwin, Master. Sidney Leonard    46.900000
164  Panula, Master. Eino Viljami       39.687500
183  Becker, Master. Richard F         39.000000
```

Which adult male passenger (age ≥ 18 and Sex is 'male') paid the highest fare relative to their class?

To answer the question perform following operations:

1. Add a column to the Titanic dataset, fare per class, containing the fare divided by the passenger class i.e. Fare / Pclass.
2. Subset rows where the passenger is male (Sex is "male") and an adult (Age is greater than or equal to 18), assigning this to adult males.
3. Sort adult males by descending fare per class, assigning this to adult males srt.
4. Select only the Name, Age, and fare per class columns of adult males sr and save the result as result.
5. Look at the result.

```
Adult male passengers with highest fare relative to class:
      Name  Age  Fare_per_Class
737  Lesurer, Mr. Gustave J  35.0      512.3292
679  Cardeza, Mr. Thomas Drake Martinez  36.0      512.3292
438      Fortune, Mr. Mark  64.0      263.0000
27  Fortune, Mr. Charles Alexander  19.0      263.0000
118  Baxter, Mr. Quigg Edmond  24.0      247.5208
```

3.3 Exploratory Data Analysis with Group-by Method Practice Exercise:

Based on the dataset Answer the following question:

What percent of the total fare revenue came from each passenger class?

To answer the question perform following operation:

1. Calculate the total Fare paid across all passengers in the Titanic dataset.
2. Subset for passengers in first class (Pclass is 1) and calculate their total fare.
3. Do the same for second class (Pclass is 2) and third class (Pclass is 3).
4. Combine the fare totals from first, second, and third classes into a list.
5. Divide the totals for each class by the overall total fare to get the proportion of fare revenue by class.

```
Total Fare Revenu: 28693.9493
First Fare Total Fares: 18177.4125
Second Fare Total Fares: 3801.8417
Third Fare Total Fares: 6714.6951

Fare totals list: [np.float64(18177.4125), np.float64(3801.8417), np.float64(6714.6951)]

Percentage of total fare revenue by classes:
First class: {proportions[0]:.2f}%
Second class: {proportions[1]:.2f}%
Third class: {proportions[2]:.2f}%
```

Based on the dataset Answer the following question:

What percent of the total number of passengers on the Titanic belonged to each age group (e.g., child, adult, senior)?

To answer the question perform following operation:

1. Create a new column, age group, that categorizes passengers into "child" (age < 18), "adult" (age 18{64), and "senior" (age 65 and above).
2. Calculate the total number of passengers on the Titanic.
3. Count the number of passengers in each age group.
4. Divide the count of each age group by the total number of passengers to get the proportion of passengers in each age group.
5. Display the proportion as a percentage.

```
Total Passengers: 891  
  
Passenger distribution by age group(%):  
AgeGroup  
Adult      86.083053  
Child      12.682379  
Senior      1.234568  
Name: count, dtype: float64
```